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Duplex Stainless Steel Laser-Surface Textured: Stability in Brine Solution

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ABSTRACT

The stability of duplex stainless steel (DSS) laser-surface textured (LST) in brine solutions is of significant interest for industrial applications. The present study reports evidences that passive films on smooth DSS2205 surfaces (Blank) deteriorate early (<96 h) if compared to oxidized and textured surfaces of LST samples (>96 h). This lower performance is detected in anodic polarization and in accelerated cyclic polarization tests with increasing exposition time and in harsh conditions. Although both surfaces maintain their chemical resistance against corrosion in noncyclic potentiodynamic experiments, the LST microstructure shows a broader passivation area with a larger potential range than the control samples. Moreover, the DSS2205 microstructure patterned surface is preserved. On the contrary, localized corrosion with macroscopic pits and microcracks is extended on the whole Blank surface. The α/γ crystalline phase ratio was reduced in corroded flat surface, while it remains stable in the corroded LST specimens, thanks to a significant formation of $FeCr_2O_4$ as a protective oxide film, as proved by Raman. Thus, LST treatments promise some advantages over nontreated interfaces, particularly for metallic pieces that will be in contact or installed close to salt environments.

1 | Introduction

Duplex stainless steels (DSSs) are Fe–Cr–Ni alloys consisting of ferrite and austenite phases with about 50%–50% ratio [1, 2]. Due to the synergistic effect between these two phases, DSS alloys exhibit higher toughness and better weldability than ferritic stainless steels, as well as higher mechanical strength and better corrosion resistance than austenitic ones [1, 2]. Their remarkable engineering performance has led to an increasing number of applications, mainly in corrosive environments such as papermaking, food processing, marine and petroleum industries, due to the combination of high mechanical strength, good toughness, and high corrosion resistance. Concerning to their intergranular corrosion, the combination of the austenite–ferrite boundary and the ferrite phase enables the precipitation of chromium carbides without severely depleting chromium at the phase boundaries [3, 4]. However, DSSs can suffer significant corrosion when implemented in fields under extremely severe service conditions [5–7]. In the last two decades, the main common causes of DSS failures in service were pitting corrosion (27%), chloride-induced stress corrosion cracking (SCC) (27%), sulfide stress cracking (18%), microbially induced corrosion (9%), selective corrosion (9%), hydrogen-induced stress cracking (5%), and crevice corrosion (5%) [5, 6, 8, 9].

The protection of DSSs against corrosive media is achieved due to the passive film formed on the top surface, which is mainly composed of mixed hydroxides and oxides [10]. In stable working conditions, the degradation and rupture process of the passive film is a continuous metal oxidation process. The film structure

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and film thickness of ferrite and austenite are very similar, whereas the chemical composition differs [11–14]. The contents of Cr, Mo, and W are higher in the ferritic passive film than that in the austenitic one, whereas the N and Ni contents in the austenitic passive film are higher than those in the ferritic one. Environment variables such as high temperature [15–17], remarkable pressure [18], mechanical vibration, high applied potentials, and aggressive environments (sulfide, sodium hydroxide, and chloride ions) usually affect the overall thickness [19–22], oxide ratio, and defect concentration of the passive film, but there are no significant changes in the typical double-layer arrangement [23].

Regarding the effect of the environment on the corrosion protection of 2205 duplex metals, hydrogen sulfide generates hydrogen ions via the reacidification effect [20]. Adding hydrogen sulfide thins the passive film, and consequently, the number of oxygen vacancies increases as well and Fe^{2+} cations are consumed. Furthermore, the influence of the applied potential on the passive film can be divided into anodic and cathodic potentials, compared with the corrosion potential (E_{corr}). At low anodic potentials, the passive film is composed of Cr_2O_3, FeO, NiO, and MoO_2, which contains a high density of cationic vacancies [17]. At high anodic potentials, the main components are Fe_2O_3, FeOOH, CrO_3, and MoO_3. Therefore, the applied potential shifts from protective to transpassive potential range, where the electron removal increases the number of the valence species and the passive film dissolves [16].

The influence of SS alloying elements on pitting corrosion mechanisms can be summarized as: difficulty of surface stabilization [24], passive barrier [25], formation of soluble precipitates [26], soluble inclusions [27], insoluble inclusions [28], and wrapping [6] mechanisms. As, for example, in the case of pitting corrosion induced by precipitates, the corrosion may be initiated due to: (i) Cr-depleted zones surrounding precipitates [29–32]; (ii) the microgalvanic effect between Cr-enriched phase and Cr-depleted phase [33–36]; (iii) the high-stress field around the precipitates [34, 37], and (iv) the α'-martensite presence induced by strain [38]. Therefore, although DSSs are more resistant to SCC than austenitic ones, they also suffer the chloride-induced SCC under certain conditions, mainly when immersed in seawater and alkaline environments [39, 40]. Then, the amount of pitting corrosion initiated in the ferrite phase is significantly larger than that in the austenite phase, according to Örnek et al. [41]. Moreover, the cracking mechanism is associated with ferrite dissolution and hydrogen embrittlement [42]. Although pitting corrosion is preferentially initiated in the ferrite phase, the cracks mainly propagate along the ferrite/ferrite boundaries and austenite/ferrite phase boundaries. In hydrogen-induced stress cracking, phase boundaries are strong hydrogen traps and hydrogen diffusion pathways [43]. In sulfide stress cracking, this phenomenon is mainly associated with the acidification effect of sulfide [44, 45].

Nowadays, different approaches have been used to modify the surface properties of DSS components, achieving new chemical, physical, morphological, and metallurgical surface properties and enhancing the corrosion resistance. For instance, Raja et al. [46] reported that DSS2205 treated at 400°C by plasma nitrocarburizing exhibited the highest corrosion resistance, along with a significant increase in surface hardness. Wang et al. [47]. improved the erosion–corrosion resistance through a nitrocarburizing treatment in a salt bath at 430°C. Majatan etal. [48] observed an enhancement of the corrosion resistance of the electrical discharge-treated 2205 DSS for biomedical applications. Lohrengel et al. [49] applied the electrochemical machining in $NaNO_3$ with very good results. On the other hand, hot-working processes such as additive manufacturing of complex shape parts and welded joints of structures produce undesired structural and microstructural changes that unavoidably deteriorate the corrosion resistance of the DSSs. To restore their corrosion resistance, several postmanufacturing treatments of the welded interfaces have been proposed, including annealing treatments [50, 51], electron beam texturing [52], laser surface heat treatment of welded interfaces [53], and combining the treatment of electron beam cladding and subsequent nitrocarburizing [54].

More recently, laser-based texturing technique have been used for the preparation of superhydrophobic surfaces in aqueous solution, which are transformed in effective-protective passive films [55]. Compared with other techniques, laser texturing presents interesting advantages such as quick processing time, simple operation, precise control of patterns, and compatibility with multiple materials [56, 57]. Some studies have reported the correlation between (super)hydrophobicity and the corrosion resistance of laser-textured stainless steels such as AISI 304 [58–60], AISI 316 [61–65], AISI 420 [66], and 17-4 PH [67]. Recently, we have explored the corrosion resistance of DSS2205 [68] and 301LN [69–71] sheets in molten carbonate at 600°C, after laser-surface texturing (LST). The laser texturing was carried out with a pulsed laser machine, selecting adequate conditions, to generate a periodic surface pattern. With this controlled surface oxidation, the corrosion rate was reduced by 48%, if compared to nontextured sheets. This substantial improvement has been attributed to the presence of oxide layers, composed of chromium oxide (Cr_2O_3), nickel oxide (NiO), and magnetite phase (Fe_3O_4), which were present at both the valleys and micropillars of the treated surface. In the present work, the novelty is to evaluate the stability of DSS 2205 (hereafter denoted DSS2205 for simplification) modified by LST, after cyclic electrochemical tests, representing the worst-case scenario for the metal protection. Therefore, a detailed analysis was performed to investigate the effects of laser texturing on the DSS2205 surface by means of electrochemical assays (potentiodynamic polarization and cyclic potentiodynamic polarization) with increasing time (life-span evaluation). The reversibility of corrosion resistance due to the interaction of LST DSS2205 surfaces with 3.5 wt.% NaCl was not investigated before. Moreover, the surface oxide wettability and microstructure analyses, before and after corrosion affectation, were evaluated by water contact angle (WCA) and scanning electron microscopy (SEM)/energy-dispersive spectroscopy (EDX) microscopy. The results were compared to smooth DSS2205 surfaces (Blank), in the same conditions.

2 | Experimental Procedure

2.1 | Materials

This study utilized DSS, specifically the 2205 grade, corresponding to UNS S31803, as the primary material. The samples, each measuring $8 \times 8 \times 2\,mm^3$, were selected for analysis. The standard chemical composition for this grade is presented in Table 1.

2.2 | Preparation of LST Samples

Before laser-surface treatment, the samples were initially ground using silicon carbide paper, followed by sequential polishing with

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TABLE 1 | Chemical composition (wt.%) of DSS 2205.

Fe	Cr	Ni	Mo	Mn	C	N	Si	P	S
Bal.	23	6.5	3.5	<2.0	<0.03	<0.2	<1.0	<0.03	<0.02

diamond suspensions of decreasing particle size down to 1 μm. Following the polishing process, the samples were subjected to ultrasonic cleaning in ethanol to thoroughly remove any residual contaminants. The laser system employed in this experiment was the Explorer-One-349-120, manufactured by Spectra-Physics (MA, USA), operating in pulsed mode. This system utilizes an Nd:glass medium, characterized by a wavelength (λ) of 349 nm, a maximum peak power of 60 W, a maximum repetition rate of 1 kHz, a focal length of 80 mm, and a spot size diameter of 145 μm.

Figure 1a illustrates a schematic representation of the spatial arrangement of the laser setup relative to the laser beam and the workpiece surface. The stand-off distance, defined as the distance between the workpiece surface and the nozzle, was set to 150 mm. The laser's focal plane position (FPP) was 70 mm from the workpiece surface.

For this experiment, a square pattern was selected for this experiment, following the strategy and processing parameters detailed in previous studies [68–70]. The laser texturing was systematically executed using WeldMark software v12, with the laser oriented at 90° to the target surface, covering the entire sample surface. Initially, vertical tracks were applied, followed by horizontal tracks, maintaining a consistent intertrack distance of 100 μm. The laser power was set at 50 W, with a scanning speed of 3.6 mm s^{-1}, a pulse duration of 2 μs, and a pulse fluence of 11.81 J cm^{-2}. These parameters resulted in an 86% overlap of laser pulses (spot beam) (Figure 1b).

2.3 | Characterization of Oxide Films

2.3.1 | SEM

Sample microstructure was examined employing a field emission scanning electron microscope (Carl Zeiss Merlin FESEM, Oberkochen, Germany) equipped with EDX (Oxford Instruments INCA-350 system, United Kingdom). Image analysis was performed on the top and cross-sectional surfaces of the samples. Focused ion beam (FIB) (Carl Zeiss Neon 40 FIB, Oberkochen, Germany) was used to obtain cross-sectional images to observe the microstructure of the duplex steel, beneath the LST surface.

2.3.2 | X-Ray Diffraction

The crystalline phases of the oxide scales were determined through X-ray diffraction (XRD, Bruker, D8-Advance, Billerica, MA, USA) using Cu Kα radiation (operated at 40 kV and 40mA). The data collection was carried out at room temperature, from 30° and 80°, with a step size of 0.01° and a collection time of 1 s/step. Phase identification was performed using the JCPDS database and the DIFFRAC Plus EVA software by Bruker AXS.

2.3.3 | Raman Spectroscopy

The location of phases was determined using Raman spectroscopy (inVia Qontor, Renishaw). Two lasers with distinct wavelengths of the applied excitation line were used in the infrared region (785 nm) and in the visible region (532 nm). An optical microscope with a 100× objective was employed to determine the analysis zone.

2.3.4 | Fourier Transform Infrared

Fourier transform infrared (FTIR) analysis was performed using a Nicolet 6700 spectrometer (Thermo Scientific) equipped with a CsI beam splitter. Measurements were conducted in reflectance mode using accessories for transmission, grazing-angle reflection, and attenuated total reflection (ATR).

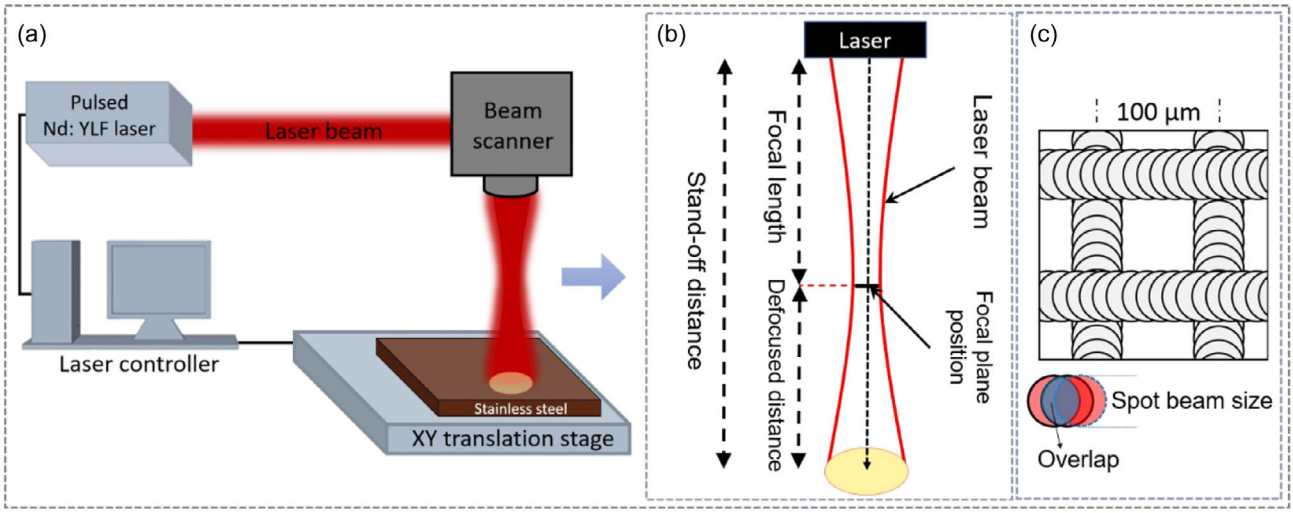


FIGURE 1 | Schematic representation of: (a) the Explorer-One 349 pulsed laser setup, (b) laser beam distances, and (c) the square-shaped pattern with spot beam size and overlap.

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2.3.5 | Surface Topography

The topography of the surfaces, with and without LST treatment, was characterized using a confocal laser optical profilometer (Olympus Lext OLS30, Tokyo, Japan) equipped with several lens objectives from ×10 to ×100 magnifications in order to assess the impact caused by the presence of chlorine ions and the potentials applied for the cyclic experiments described in the next section.

2.4 | Electrochemical Assays: Corrosion Evaluation

The potentiodynamic polarization experiments on LST and Blank samples were conducted using an Autolab PGSTAT302N Galvanostat equipped with the ECD module (Ecochimie, The Netherlands). A three-electrode setup was employed in a one-compartment cell containing 200 mL of naturally aerated 3.5% NaCl solution, as the supporting electrolyte. The temperature and pH were only monitored at the beginning and at the end of 1-week experiments. The temperature did not vary from 23 to 25±1°C in the period of analysis and the pH decreased from 7.5 to 8.0, at beginning of test, to acidic values (∼6.0), after 1 week (measured with pH test strips). DSS samples, both laser-treated and untextured (Blank), served as the working electrodes (WE) and the area of analysis was delimited by the cell hole (0.1963 cm^2), adapted with a rubber gasket for sealing. A platinum wire was used as the counter electrode (CE), and an Agl AgCl electrode containing KCl 3 M aqueous solution acted as the reference electrode (RE) ($E°$ = 0.197 V at 25°C vs. SHE). Polarization analyses were carried out by measuring linear sweep voltammetry (LSV) scans, using a negative potential starting at −1.0 V and raising to positive values (vs. E_{OCP}) and with a scan rate of 1 mV s^{-1}. The system was stabilized for 30 min before recording the open-circuit potential (E_{OCP}), based on previous works reported [72, 73].

The same conditions were used to perform the cyclic polarization (CP) analyses to evaluate the pitting corrosion resistances of both sample types. In this case, two LSV scans were performed to create a cyclic polarization. The first scan started at a potential of −0.5 V and progressed to 1.5 V, while the second scan immediately followed, reversing from 1.5 V back to −0.5 V. The WE area in this test was larger than in the potentiodynamic anodic polarization assays (0.7854 cm^2) in order to be able to evaluate pit and crevices formations and the chemical modification of oxide layer by SEM–EDX and Raman analyses in a broader zone. The scan rate was set to 1mV s^{-1}. As-prepared LST samples were used for this analysis. Several measurements were conducted, with controlled intervals, to monitor corrosion behavior over a 7-day period. The electrochemical cell, the electrodes, and the electrolyte solution were the same as described above. Prior to electrochemical measurements, the samples were immersed in the electrolyte for 30 min to obtain a steady state (E_{OCP}), based on the previous OCP stabilization time achieved with potentiodynamic polarization experiments described above [72, 73].

2.5 | Surface Wettability

The static WCA measurements were conducted using a Krüss Drop Shape Analyzer goniometer to evaluate the wettability of the samples. The instrument was calibrated and utilized under standard laboratory conditions at room temperature. For each sample, ten measurements were taken to ensure accuracy and repeatability. The water drop (volume of 0.5 µL) was deposited in the delimited zones serving as working area for corrosion tests. Variations on the WCA values were used to follow up the surface wettability of the abraded samples before and after corrosion assays. 3D topography maps of untreated and laser textured surfaces, prior and after the deposition of one droplet, were taken with a confocal microscope Zeta-20, from KLA microscope.

3 | Results and Discussion

3.1 | Microstructure and Passivating Layer of DSS2205 LST Samples

3.1.1 | Microstructural and Topographical Analysis

After LST treatment, the surface composition and microstructure of the samples were analyzed to assess the effects of LST in detail. Figure 2 displays the SEM images of both the Blank and LST-treated microstructures. The Blank sample displayed a completely flat surface post-polishing (Figure 2a,b), while the LST-treated sample exhibited characteristic groove patterns and periodic wave formations induced by the laser (Figure 2c,f). The images indicate that the grooves measured approximately 70 µm in width between the valleys of the laser tracks, with each valley measuring around 30 µm in width. As illustrated in Figure 2e,f, the topography images of the LST sample reveal a microstructure characterized by uniformly distributed, square-shaped micropillars without pronounced grooves and valleys. Additionally, it is crucial to highlight the consistent roughness observed across both the valleys and the micropillars, which are adorned with redeposited material from the laser track. This favorable result is attributed to the carefully optimized laser parameters, including scanning speed, pulse duration, and pulse fluence, all optimized for the laser–metal interaction in the DSS2205 alloy.

Surface topography was further characterized using confocal profilometry to confirm the efficacy of cleaning and texturing processes. Figure 3 displays the 3D reconstructed images of the Blank surface and the LST pattern. These images illustrate that laser processing significantly increases surface roughness by incorporating textural features. The width of the valleys was approximately 30 µm, while the height of the valleys between the micropillars ranged from 3 to 5 µm. The width dimensions of the grooves and pillars obtained from the surface profile are similar to those observed in the SEM images.

3.1.2 | Chemical Composition and Structural Analysis of Oxide Phase Distribution

The phases present on the surface of LTS sample have been identified by XRD. Figure 4a shows the XRD spectra of the Blank and LST samples. In the spectrum of the LST sample, the LST spectrum revealed prominent peaks corresponding to the austenite and ferrite phases of the DSS2205. In addition, minor peaks of magnetite (Fe_3O_4) were detected, which agrees with the presence of oxygen observed in the SEM–EDX composition analysis. Raman spectroscopy was used to locally analyze the chemical composition of both groove and pillar regions. In Figure 4b, the main bands of Raman spectra in the region of 200–800 cm^{-1} corresponded to the magnetite phase (Fe_3O_4), with the most intense

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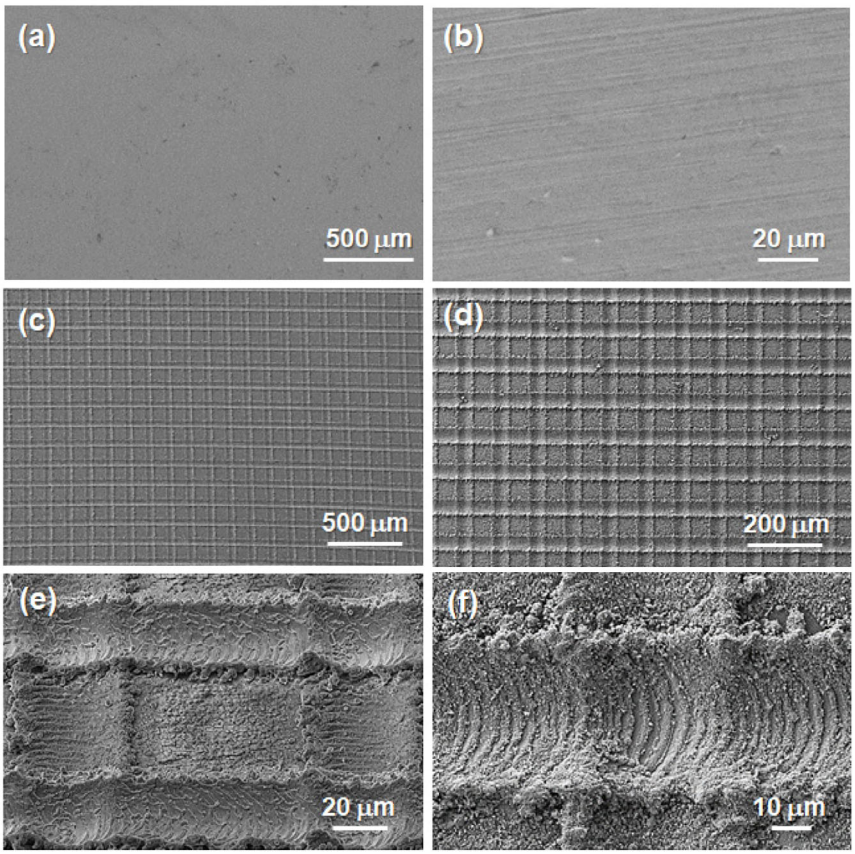


FIGURE 2 | SEM micrographs at different magnifications of: (a,b) Blank surface showing the smooth texture and some marks after grinding process and (c–f) LST DSS2205 micropatterned surface, produced after laser-surface treatment. Low magnifications (a–d); high magnifications (e,f).

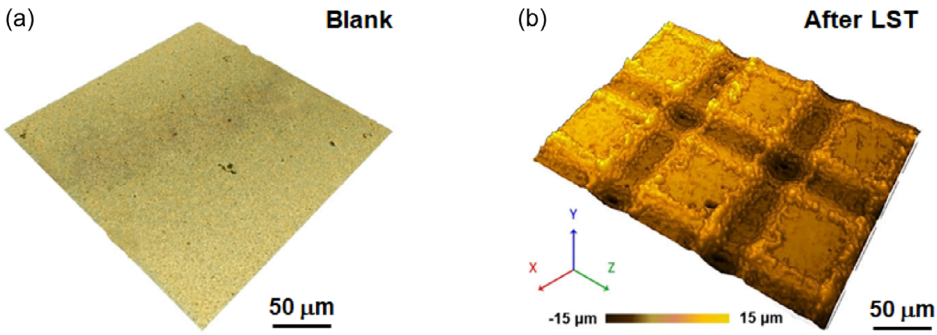


FIGURE 3 | 3D reconstruction of a representative: (a) Blank smooth surface and (b) LST textured surface, obtained using confocal laser optical profilometry.

band at 690 cm^{-1} [74], and lower amount of chromium oxide (Cr_2O_3) at 610 cm^{-1} [75]. The Raman spectra presented a similar intensity at both the grooves and pillars, which confirms the XRD results and suggests that the LST treatment generated a homogeneous phase distribution in the entire surface of DSS2205. In the Raman spectra, between 1000 and 1800 cm^{-1} (Figure 4c), two broad bands centered at ~1350 and ~1560 cm^{-1} are observed, which are attributed to the D and G carbon characteristic modes, respectively. The peak at ~1350 cm^{-1}, the so-called D band, arises from disorder and defects such as edges, vacancies, or functional groups, in the sp^2 carbon network for activation [76]. The peak at ~1560 cm^{-1} (G band) is associated with the in-plane vibrational mode (E_{2g} symmetry) of sp^2-hybridized carbon atoms arranged in a two-dimensional hexagonal lattice, such as graphite or graphene [76]. Considering the characteristics of the D and G peaks such as their relative intensity (I(D)/I(G) ratio ≈ 1), the shift of the G peak to low frequencies (~1560 cm^{-1}) within the range of ~1550 –~1590 cm^{-1} and the broad width of D peak (FWHM) led us to conclude that the carbon layer is predominantly present in amorphous or highly disordered graphitic state, in both the grooves and pillars.

FTIR analyses were conducted to qualitatively assess the presence of chemisorbed organic compounds on both the LST and Blank surfaces after they were cleaned and maintained in a laboratory environment for 30 days. As illustrated in Figure S1, the LST surface demonstrated a markedly higher level of chemisorption of

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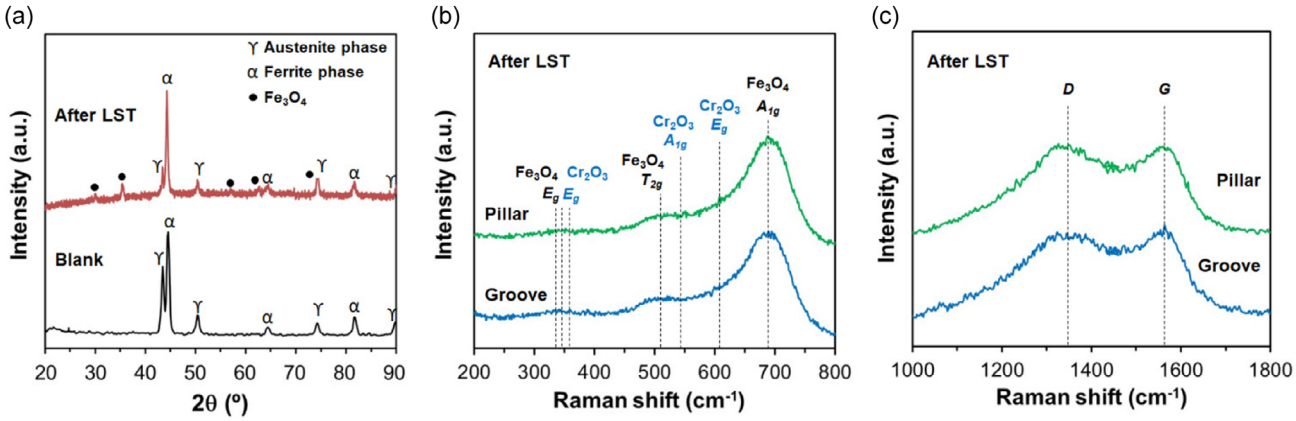


FIGURE 4 | (a) XRD spectra of the Blank and LST surfaces. (b,c) Raman spectra of the Blank and LST surfaces within the regions of: (b) 200 and 800 cm^{-1} and (c) 1000 and 1800 cm^{-1}, using two distinct lasers in the visible region (532 nm) and in the infrared region (785 nm), respectively.

carbon-containing species, evident from the pronounced intensity of infrared bands corresponding to several functional groups, including alkanes, alkenes, carbonyls, and hydroxyls, among others. These findings indicate that the presence of chemisorbed organic compounds on the LST surface may significantly influence the formation of the oxide layer and the corrosion mechanisms. This issue will be further explored in the subsequent sections.

3.2 | Corrosion Resistance in Salt Solution and Surface Characterization

3.2.1 | Potentiodynamic Polarization Curves

Having proved that the DSS2205 LST samples were well patterned and protected by passivating oxide layers, corrosion tests were performed in concentrated salt solution to monitor the evolution of such film stability. First, the flat surface of sheets without laser texturing was subjected to polarization assays during 1 week. Figure 5a shows the potentiodynamic polarization curves of Blank samples, before and after their immersion in 3.5 wt.% of NaCl solution. There is no typical Tafel region during anodic polarization due to the occurrence of transformation from activation to passivation. Therefore, in the anodic curve, a clear Flade potential (E_f) at −0.43 V is observed in the first 6 h of immersion time, in which the current density reverses and moves to lower values (7.6×10^{-7}A). The electrode surface was transformed from active to passive state, with critical passivation current density (i_{cp}) starting at 5.2×10^{-7}A cm^{-2} (Table 2). Moreover, the material presents a high range of concentration polarization in the cathodic region, from −1.0 to −0.5 V. The formation of well-known passive layers composed of Cr_2O_3, Fe_2O_3, Fe_3O_4, $Fe_2(MoO_4)_3$, and NiO above the SS surface helps to maintain the corrosion current density (i_{corr}) at low values (μA cm^{-2}) with increasing immersion time (Table 2, above 6 h) [77]. However, the corrosion potentials (E_{corr}) move to positive values and the anodic curve slowly begins to lean toward the area of positive current densities, with increasing potential (range from −0.3 to +0.3 V in Figure 5a), from 72 to 168 h of salt solution

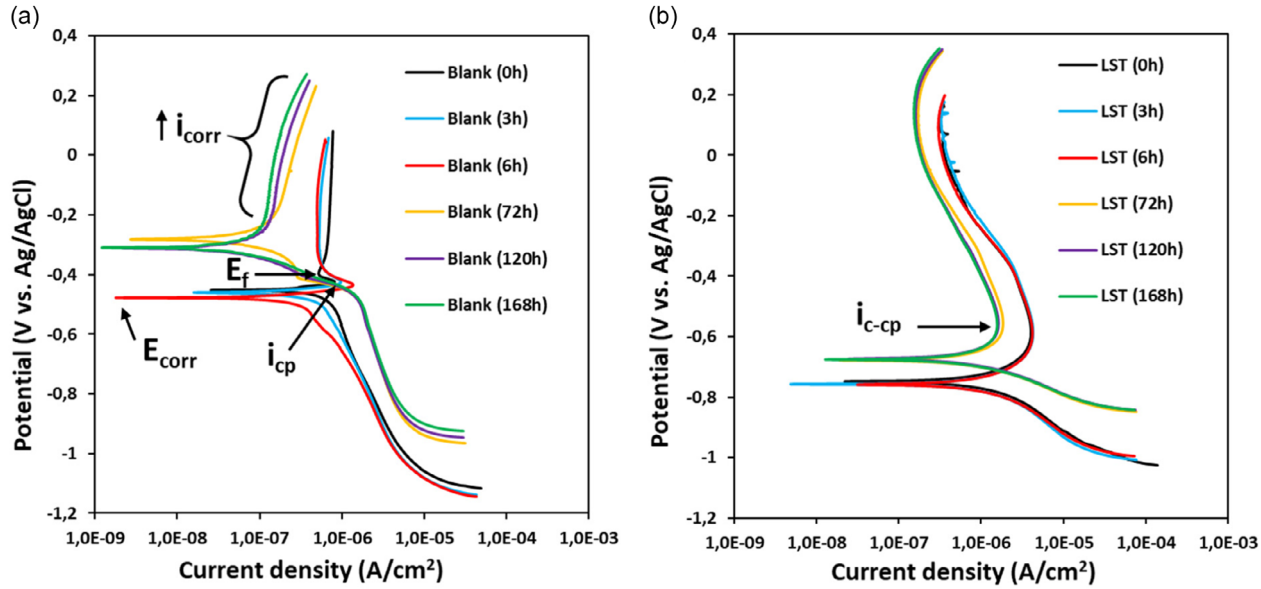


FIGURE 5 | Potentiodynamic anodic polarization curves of: (a) Blank samples and (b) LST samples, with different intervals of immersion time, in NaCl 3.5 wt.%. Labels highlighted in plots correspond to: i_{corr}, corrosion current density; i_{cp}, critical passivation current density; E_{corr}, corrosion potential; E_f, Flade potential; and i_{c-cp}, pseudo-critical passivation current.

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TABLE 2 | Characteristic parameters obtained for Blank and patterned DSS2205 samples in NaCl 3.5 wt.%, with increasing exposure time, under potentiodynamic anodic polarization.

Blank samples	**0 h**	**3 h**	**6 h**	**72 h**	**120 h**	**168 h**
E_{OCP}[a], V	−0.12	−0.14	−0.15	0.03	0.05	0.07
E_{corr}[b], V	−0.45	−0.46	−0.48	−0.28	−0.31	−0.31
I_{corr}[b], A cm^{-2}	9.73×10^{-7}	5.97×10^{-7}	4.17×10^{-7}	7.86×10^{-8}	1.72×10^{-8}	1.77×10^{-8}
LST samples	**0 h**	**3 h**	**6 h**	**72 h**	**120 h**	**168 h**
E_{OCP}[a], V	−0.03	−0.02	−0.002	0.14	0.15	0.15
E_{corr}[b], V	−0.75	−0.76	−0.76	−0.67	−0.67	−0.68
I_{corr}[b], A cm^{-2}	2.53×10^{-7}	1.97×10^{-7}	2.54×10^{-7}	1.63×10^{-7}	1.48×10^{-7}	1.75×10^{-7}
$I_{p\text{-}cp}$, A cm^{-2}	4.20×10^{-6}	4.20×10^{-6}	4.20×10^{-6}	1.58×10^{-6}	1.56×10^{-6}	1.56×10^{-6}

[a]E_{OCP} measured after 30 min of sample immersion in NaCl solution.
[b]Data obtained by extrapolation method with the cathodic curve and E_{corr} intersection.

immersion. The corrosion potentials (E_{corr}) and the corrosion current densities (i_{corr}) of the polarization curves were obtained by extrapolation analysis, by the intersection of cathodic polarization curves and the corrosion potential.

In our previous studies [68], it has been proved that LST DSS2205 samples have thicker nonpassive and passive oxide layers composed mainly of Fe_3O_4, Cr_2O_3, NiO, and amorphous or highly disordered graphitic carbon (C), distributed by layers. When studying the LST samples in contact with NaCl solutions, at the same conditions than Blank (Figure 5a), a clear activation to positive current densities is observed for the first case, i.e., in textured sample (Figure 5b). The E_{corr} is much more negative than in the Blank plates, as expected, since the LST surface is already oxidized by the controlled laser treatment. Additionally, this steady layer provokes the appearance of a much larger passive range (from −0.60 to +0.07 V) than that observed for the Blank sheets, without any apparent Flade potentials. However, a pseudo-critical passivation current ($i_{p\text{-}cp}$) was determined when the current value goes back into the anodic zone (Figure 5b), with progressive current depletion from 6 to 72 h of polarization test. Moreover, after 3 days of contact with the brine solution, the LST surface remains highly stable until 1 week, with no variation either in the E_{corr} or in the i_{corr} data (Table 2). Similarly, the i_{corr} was obtained by extrapolation analysis due to the absence of Tafel behavior. The values are expressed in Table 2 for comparison with the Blank samples.

As explained in previous sections, the laser energy applied (11.81 J cm^{-2}) and the scanning speed (3.6 mm s^{-1}) chosen do not exceed the heating limits that could cause an excessive melting of the passive layer of DSS2205, as observed in other studies [78]. Thus, the compact structure of protective oxides is the responsible for the large passivated zone found in those samples. Detailed inspection of SEM images also proved that the textured and rough topography obtained with the nanosecond-pulsed laser treatment is mostly preserved after 7 days of corrosion analysis, corroborating with the high stability of this oxide layer (Section 3.2.2).

3.2.2 | Topography Analysis and Chemical Composition after Potentiodynamic Corrosion Tests

Comparison of SEM images before and after potentiodynamic polarization tests led us to confirm that both, textured and Blank surfaces, demonstrated resistance to corrosion when exposed to a chloride solution for prolonged time (7 days). In Figure 6a, the marks of surface grinding before analysis are visualized, together with some punctual defects. Similarly, comparable lines are evident after 1 week of contact with the NaCl solution, together with localized holes, probably due to defects on sample grinding (Figure 6b), and some soluble precipitates or inclusions above the surface and close to holes. Xiao et al. [72] have described the micromorphology of corrosion pits in DSS2205 in contact with 35 g L^{-1} NaCl solution, at different pH values, and after polarization tests. They found a characteristic funnel-shaped form, shallow, with small and medium sizes, to some extent. Therefore, based on the micrography analysis, it can be affirmed that the metal integrity of the DSS2205 passive protective surface was apparently preserved against pitting corrosion over 1 week of anodic polarization testing.

In LST sheets, the micropatterned morphology observed by SEM corresponds to the microstructure already described in Section 3.1 and also in our previous study [68], with the typical groove marks with periodic waves on valleys generated through the laser treatment (Figure 6c,d). Unlike Blank samples, before and after salt solution immersion for 1 week LST surface does not present any other marks than that related to the laser treatment (Figure 6c–f, respectively). However, the waves observed on valleys for the as-prepared samples have slightly changed, which can indicate some local affectation on the density of the protective oxide layer. Nevertheless, this phenomenon was observed only in a few zones of analysis. Deeper evaluation of the oxide presence in such grooves and in scan-line separations was performed with SEM–EDX mapping to discard any enhancement of corrosive product provoked by the salt solution and the electrochemical perturbation.

The smooth and uniform surface of Blank sample before corrosion test (Figure 7a) was checked, and some localized microscopic defects were visualized again in the SEM imaging used for mapping exploration (Figure 7b), similar to the precipitates indicated with arrows in Figure 7b. The rectangular zone marked in both images corresponds to the zone chosen for the identification of the changes in the chemical composition of the main alloying elements. The EDX analysis shows a practically invariable composition of Fe, Cr, Ni, Mo, and Mn, before and after the corrosion tests, for all samples (Table S1), confirming the high

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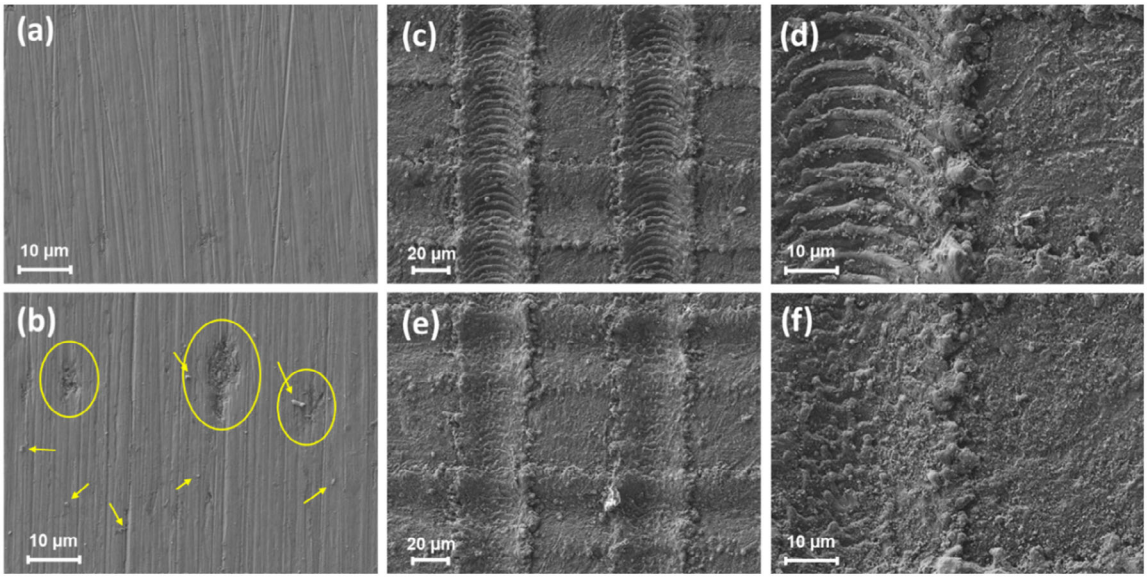


FIGURE 6 | SEM micrographs of: (a) Blank sample before polarization; (b) after corrosion tests; (c,d) LST samples, at different magnifications before polarization; and (e,f) after corrosion polarization tests in NaCl 3.5 wt.%. Yellow circles and arrows in (b) indicate the holes and the precipitates, respectively, at the surface of bare DSS2205 sample, which appeared after 7 days of immersion in NaCl 3.5 wt.%.

performance of this DSS in chlorine environments. The EDX mappings of non-LST sheets (Blank) and laser-textured (SLT) evidence the presence of all main alloying elements and slight atoms (C and O), before and after immersion in salt solution for 7 days. The largest difference in composition remains in the amount of C and O in the LST sheets, which is topographically higher than in Blank flat surfaces due to the oxidation process caused by the laser treatment and, subsequently, the passivation process. Such atoms are present in pillars and grooves. On the contrary, it is important to emphasize that the Cr/Fe and Cr/Ni ratios remained high for both typology of surfaces. The high Cr/Fe ratio, combined with high O content, in the LST surface indicate that the corrosion layer is thicker than that formed on the Blank surface. The EDX spectra of Blank and LST samples are presented in Figures S2 and S3.

The EDX mapping (Figure 7a,b) offers one visual evidence of the presence of most relevant metals and nonmetallic components above the alloying surface. There is one apparent reduction of metals such as Mo when comparing the images before and after assays (confirmed by the EDX weight percentage reduction in Table S1), and localized increasing of oxygen concentration in the agglomerated particles revealed in SEM micrograph of Blank after polarization test. In SEM-mapping images of LST samples, the grooves observed after 7 days of NaCl immersion (Figure 7c,d) have apparently the same line marks and micropillars of LST DSS2205 before corrosion, proving the high stability of the oxide passivating layers in front of aggressive agents such as chlorine ions. When analyzing the mapping images on the right (Figure 7d), no signs of pitting corrosion were detected since the Fe content is invariable and oxygen atoms are spread over the entire zone. Moreover, Cr/Fe and Cr/Ni ratios remain almost unaltered in both cases, before and after analysis (Table S1). This indicates that the chemical composition of the oxide phases formed during the corrosion process persisted largely unchanged. Several works have reported the superior corrosion resistance of metals with controlled passivating films formed after laser surface ablation treatments [78–81]. Furthermore, the conditions applied to the system (beam direction, laser distance, etc.) are sometimes the key factors that enlarge the durability of such stable films. In this work, a good balance of laser surface oxidation parameters was achieved to maintain stable the passivating layers, composed of Cr_2O_3, Fe_3O_4, NiO, and Mo_2O_3 in DSS2205 substrates, against corrosion. Nevertheless, such protective layers cannot withstand long periods of contact with a brine environment or more aggressive media. In such conditions, pitting and crevice corrosion can appear. Therefore, one of the main objectives of this study was to evaluate these phenomena under forced cyclic tests, as shown in the following section.

3.2.3 | Accelerated Corrosion Induced by Cyclic Polarization Tests in Salt Solution and Evaluation of Pitting Formation

The cyclic potentiodynamic polarization (CP) can be regarded as an accelerating test for rapid evaluation of pitting corrosion. The chloride-ion concentration is a main factor for pitting initiation in stainless steels. In the present study, successive CP was applied to the Blank and LST surfaces to eliminate the passive and protective oxide layers and achieve the highest pitting phenomena over time in salt water with high salinity concentration (3.5 wt.%). For this test, a larger area of the working electrode was chosen if compared to the single anodic polarization experiments, previously described.

In Figure 8a, the comparison of forward and reverse curves for Blank and LST specimens, at 0 h (first CP) and after 24 h, is shown. The potential at which the current sharply enhances is defined as the pitting potential (E_{pit}). As can be seen, in the forward line the current increases suddenly over 0.55 V for Blank sample, corresponding to its E_{pit} (Table S2). In the reverse curve, some metal dissolution, highlighted with a discontinuous red circle in Figure 8a at around +0.14 V, was observed at higher current densities than LST sample. Afterward, a repassivation potential (E_{rp} at −0.14 V), which crosses the forward scan in the anodic

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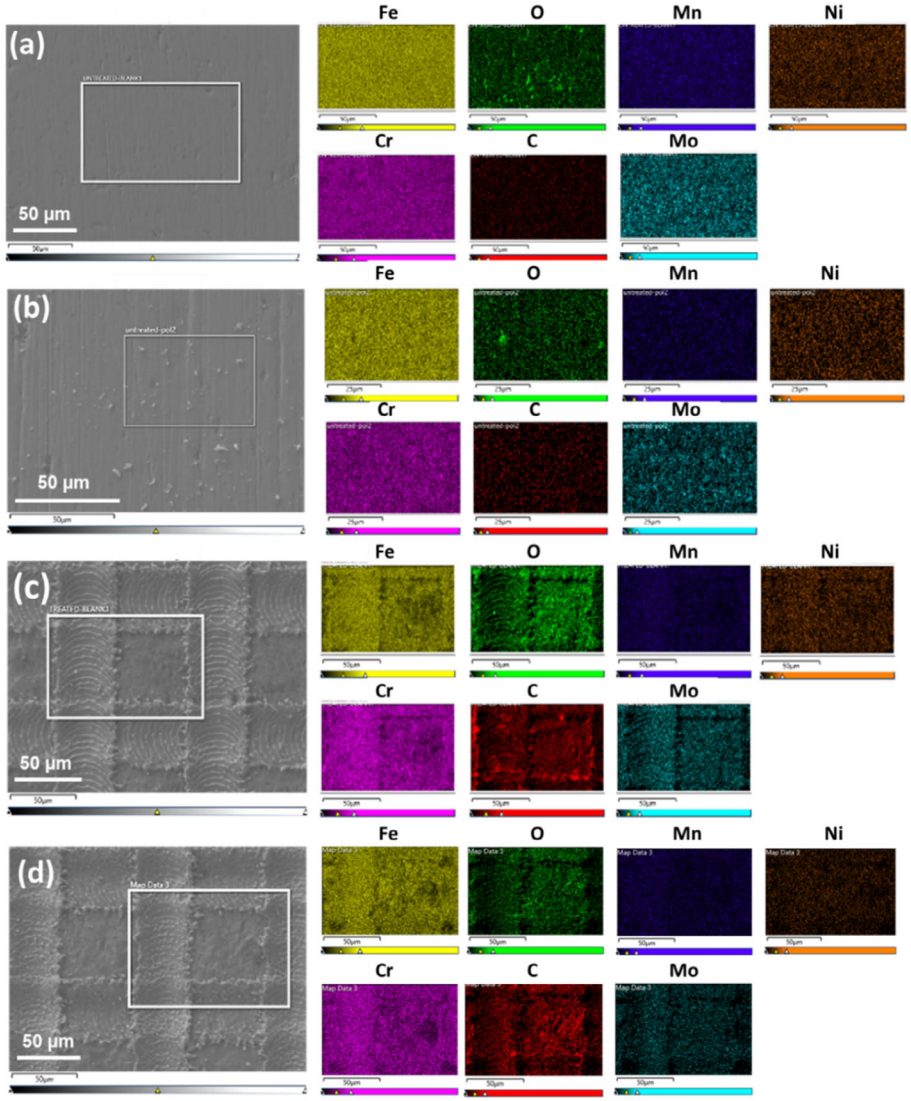


FIGURE 7 | SEM micrographs (on left) corresponding to the surfaces used for mapping analysis: (a,c) before and (b,d) after immersion of substrates in NaCl solution for 7 days; and EDX mapping images (on right) of main DSS2205 elements and alloying atoms (Fe, O, Mn, Ni, Cr, C, Mo). Micrographs (a,b) correspond to Blank sample and (c,d) to LST samples.

zone, was reached. The metal surface is prompt to pit because the hysteresis loop is large and the E_{rp} is much more negative than E_{pit} and closest to the E_{corr}. After 24 h of sheet immersion and 2nd CP test over the same area, the tendency is the dramatic reduction of E_{pit} (0.034 V) and the obtaining of increasing localized corrosion points (Blank, 24 h). Likewise, the LST surface showed a typical active behavior at first CP with lower E_{pit} (~0.33 V), with an obvious unstable pitting zone (from −0.6 to 1.1 V), with no repassivation potential in the reverse scan (Figure 8a, continuous circle marked for 0 h) and much larger hysteresis loop than Blank sample. These results indicate that pitting corrosion occurred in an early stage for both samples, which is expected due to the highly concentrated brackish solution (NaCl 3.5 wt.%) used, comparable to seawater salinity. However, small differences between micropatterned surfaces and the flat ones can be observed. The corrosion potential (E_{corr}) and the switching potential (E_{sw}) is much higher in LST treated surfaces than in Blank at initial stage, indicating a higher stability to localized corrosion attack. Furthermore, with an increasing immersion time (24 h), a similar trend to localized corrosion has been observed but the corrosion current density (i_{corr}), obtained via the extrapolating method confirms a much less positive value after LST treatment (5.48×10^{-7}A cm^{-2} compared to 2.65×10^{-3}A cm^{-2} of Blank) (Table S2). With prolonged exposition (96 h, Figure 8b), despite a similar pitting trend of both samples with wide hysteresis loops is observed, Blank sample still has a higher i_{corr} than textured specimens (1.26×10^{-6}A cm^{-2} and 7.79×10^{-7}A cm^{-2}, respectively). After 7 days (168 h) (Figure 8c), both samples show equal behavior and are unable to stand any passivating protective layer. Table S2 summarizes the evolution of E_{corr}, i_{corr}, and E_{pit}, among other parameters, and Figure S4 exemplifies the overlapped forward and reverse curves with time.

Undoubtedly, cyclability applied potentials deteriorate the protective passivating layers of DSS2205, even if the metal interface tries to repassivate, as proved with nontextured sample. The stability of DSS2205 oxides coating, composed mainly by iron oxides in the LST sample, such as magnetite (Fe_3O_4) and iron chromite ($FeCr_2O_4$), is strongly decreased under highly concentrated salty

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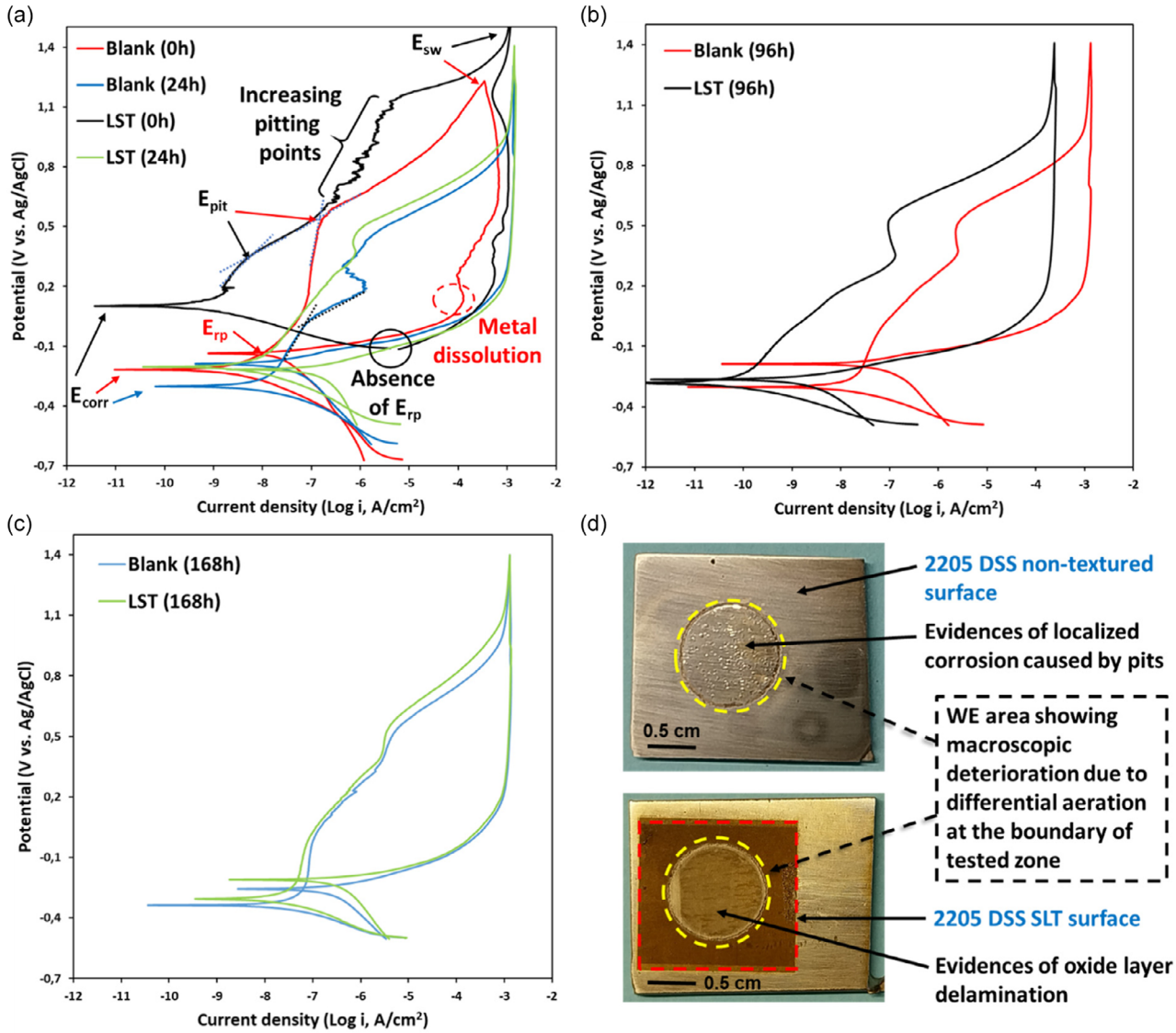


FIGURE 8 | Potentiodynamic cyclic polarization curves of: (a) Blank and LST samples (0 h and after 24 h); (b) Blank and LST samples (96 h); and (c) Blank and LST samples (168 h), in 3.5 wt.% of NaCl solution. (d) Digital photographs of Blank (up) and LST (down) sheets showing the visible deterioration of both samples. Scale bar: 0.5 cm. Labels highlighted in plots correspond to: E_{corr}, corrosion potential; E_{pit}, pitting potential; E_{sw}, switching potential; and E_{rp}, repassivation potential.

solutions (discussed in next section). The presence of macroscopic pits in Blank and the delamination of oxide layers in LST sheets can be seen in Figure 8d. However, such results can only be supported with additional microscopy and chemical structure analysis to determine the real effect of pitting in the surface chemical composition after the cyclic experiments. Therefore, SEM mapping, EDX analyses, XRD, and Raman spectroscopy were also carried out to evaluate the degree of metal damage on both typologies of surfaces: flat (Blank) and grooved (LST), after cyclic experiments.

3.2.4 | Topography Analysis and Chemical Composition after Cyclic Potentiodynamic Corrosion Tests

Duplex steels are formed by austenite (γ) and ferrite (α) and the pitting corrosion resistance of both phases depends on several factors, such as the concentration and composition of salt solution, and also on the heat treatments the steel has undergone, i.e., aging or annealing treatment [82]. Pan et al. [73] have reported that such phases change when different polarization potentials are applied to polished bare surfaces. They proved that SCC is promoted by potential shift to very low values (−850 and −1200 mV), with anodic dissolution of the metallic elements (Fe, Cr, Ni, and Mo), followed by a synergic enrichment of hydrogen in austenite. Combining SEM and EBSD, authors showed that some small pits appear on the γ-phase and become SCC nucleation sites.

In a similar way, crevice corrosion in the Blank DSS2205 sample exposed to the cyclic potentiodynamic polarization was detected by comparing the SEM images and EDX mapping, before and after corrosion assays (Figure 9a,b). Large cracks were evidenced surrounding some of the micropits, corroborating the fact that DSS2205 cannot resist such aggressive tests. Microscopic pits and crevices on bare samples were also visible by optical micrography (Figure S5a, b). On the contrary, a redistribution of oxides

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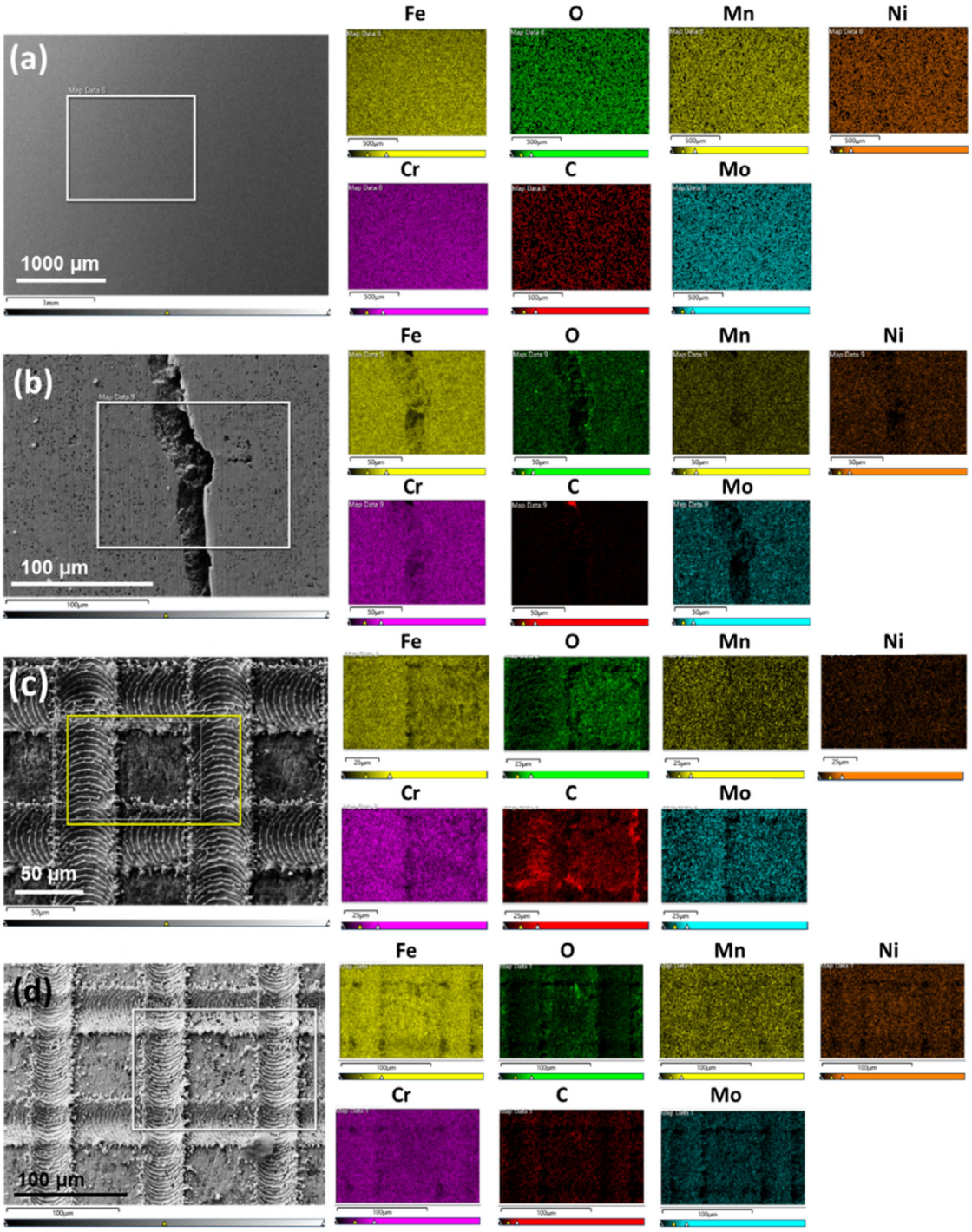


FIGURE 9 | SEM micrographs (on left) corresponding to the surfaces used for mapping analysis: (a,c) before and (b,d) after cyclic polarization tests with immersion of substrates in NaCl solution for 7 days. EDX mapping images (on right) showing the distribution composition of main DSS2205 elements and alloying atoms (Fe, O, Mn, Ni, Cr, C, Mo). Micrographs (a,b) correspond to Blank sample and (c, d) to LST samples.

was observed (by topography brightness and localized brown spots) in the LST sheet (Figure S5c,d). Surprisingly, although the textured surface (Figure 9c) showed the presence of several micropits too, mainly localized in pillars (Figures 9d and S6), the microstructure formed after laser ablation could avoid the severe damage caused by crevice corrosion. As can be seen in Table S3, comparing the samples after immersion, the LST samples exhibited a higher content in Fe, Cr, and Ni, maintaining almost constant the Cr/Fe and Cr/Ni ratios.

To complement the observations obtained from the topography SEM images showed in Figure 9, EDX mapping of cross-sectional zones and the corresponding EDX maps were analyzed for both Blank and LST-treated DSS2205 samples, after cyclic corrosion tests (Figure 10). The results revealed clear differences between the two samples. As shown in Figure 10a, the Blank sample exhibits a heterogeneous microstructure with high concentration of pores and voids as well as a pronounced elemental segregation, particularly of Mo and Ni, along with strong O signal that reflect

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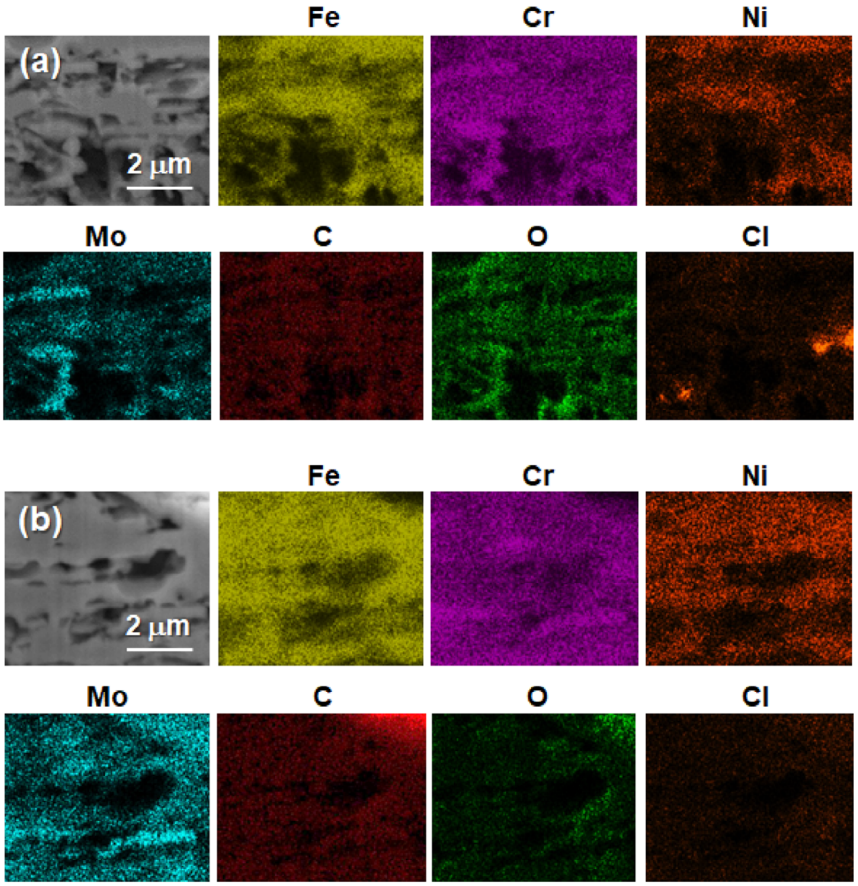


FIGURE 10 | Cross-sectional FIB-SEM micrographs of: (a) Blank and (b) LST samples. The EDX mapping images (colored images) show the distribution composition of main DSS2205 elements of alloying atoms (Fe, Cr, Ni, Mo, O, C) and Cl, after cyclic corrosion tests.

the presence of metal oxides, corrosion products, and other contaminants. Localized Cl enrichment is also detected, indicating a higher susceptibility to penetration of chloride in the early stages of localized corrosion. In contrast, the LST-treated sample showed a more uniform microstructure (Figure 10b). Elemental maps of alloying atoms (Fe, Cr, Ni, and Mo) display a significant more homogeneous distribution during corrosion test. The reduced O signal suggested a more compact and stable oxide layer formed upon cyclic potentiodynamic polarization assays over 7 days. Additionally, the Cl signal is minimal and more diffuse, indicating that LST effectively limits chloride penetration into the inner oxide layer, thereby enhancing the alloy's resistance to localized corrosion. This may suggest a significant enhancement in the formation of chromium, nickel, molybdenum and iron oxides, as a result of the LST process, indicating an alloy stabilization during corrosion assays. As described in our previous work [68], the superior corrosion resistance of LST surfaces with respect to nontextured substrates is due to the beneficial presence of a stable oxide layer, which are able to retain elements as Cr, Mo, and Ni during the corrosion process, and decreasing the diffusion of ions through oxide layer. However, these results can have a certain error associated with the quantification of light elements by EDX, such as C and O, and therefore, the C and O compounds were complementarily analyzed by XRD and Raman spectroscopy.

As shown in Figure 11a, the XRD spectra of LST and Blank samples after cyclic potentiodynamic test exhibits the characteristic patterns of the duplex steel phases (α and γ). It is important to note the remarkable differences in relative intensity of the α (~43°) and γ (~44°) peaks for the corroded Blank compared with both the Blank and LST samples before corrosion tests. This indicates that ferrite (α) was predominantly dissolved as a result of a passive film exhibiting deficient corrosion protection when exposed to severe cyclic potentiodynamic conditions. By contrast, the spectrum of the corroded LST sample was very similar to that of the blank, before corrosion, which suggests that both austenite and ferrite in that sample present comparable corrosion behavior in such aggressive test. Moreover, nonprotective iron oxides such as Fe_3O_4, observed in the LST surface before corrosion, have been apparently removed after the prolonged assay. Other metal oxides combinations were not discerned with XRD analysis and a more detailed investigation with micro-Raman spectroscope was necessary.

Raman spectroscopy confirmed the main oxide phases observed in XRD analysis, after cyclic potentiodynamic corrosion tests and immersion of samples for 7 days in the brine solution. As can be seen in Figure 11b, Raman spectra performed in the grooves and pillars of LST sample exhibit an intense and wide peak at around 650 cm^{-1}, which is attributed to the main characteristic band of $FeCr_2O_4$ passivating layer [83]. The spectra also show low intensity at the main absorption bands of NiO (520 cm^{-1}) [84], Cr_2O_3 (610 cm^{-1}) [75], Fe_3O_4 (690 cm^{-1}) [74], and MoO_3 (820 cm^{-1}) [85], discarding significant amounts of these oxides. In the Blank sample, diverse Raman spectra were obtained depending on the

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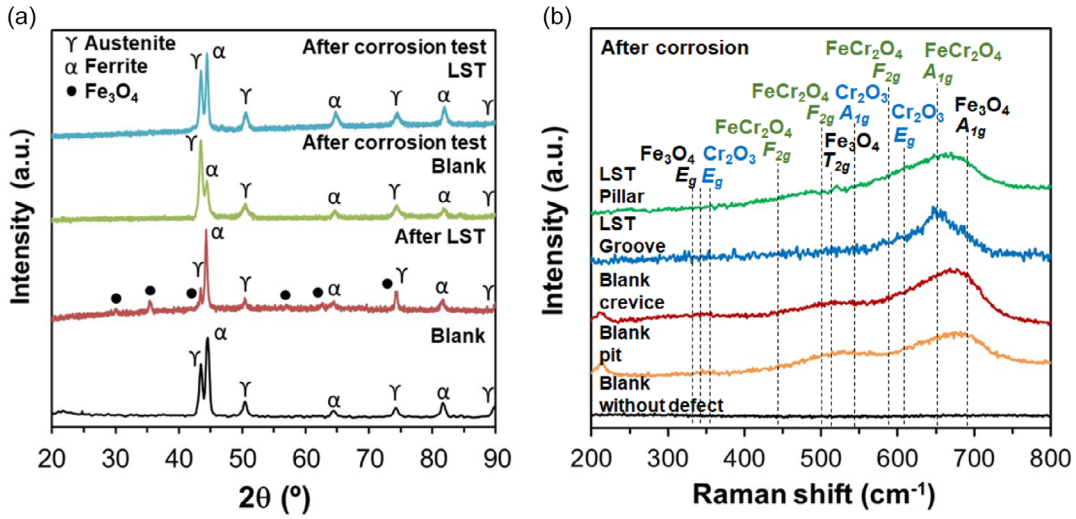


FIGURE 11 | (a) XRD spectra of the Blank and LST surfaces before and after corrosion tests. (b) Raman spectra at different regions of the Blank and LST surfaces using the laser in the visible region (532 nm).

zone of analysis, which is in a good agreement with the heterogeneous surface generated during corrosion tests. For instance, the spectra corresponding to the pits and scratch zones present the main peak of their absorption bands at ~670 cm^{-1}, which may be attributed to a mixture of $FeCr_2O_4$ (650 cm^{-1}) and Fe_3O_4 (690 cm^{-1}). In contrast, no presence of oxides was detected in the defect-free and bright metallic regions of the sample.

3.3 | Influence of Surface Wettability in the Corrosion Protection

It is widely known that the wettability of a metal surface is connected with its micromorphology. In laser-induced surface textures it depends on the widths of microgrooves [57]. In the present study, the patterning model (microgrooves width and height) was preserved by the fixed conditions detailed in Section 2.2. The WCA measurements showed that the laser-treated samples exhibited significantly higher hydrophobicity compared to the Blank samples (Figure 12a). The LST substrate before polarization tests is superhydrophobic, with WCA values higher than 150° [65]. This enhanced hydrophobicity has been extensively investigated and it is attributed to the microscale surface roughness induced by the laser treatment [60, 86]. The laser etching process creates a grooved pattern that alters the surface energy [65, 87]. Surface roughness can amplify the intrinsic hydrophobic properties by trapping air pockets and reducing the actual contact area between the water droplets and the surface. Consequently, water droplets are more likely to bead up rather than spread out on the microtextured surface, resulting in higher contact angles (i.e., an increased hydrophobicity). Thus, representing an appropriate way to delay any entrance of the aggressive electrolyte to the base steel [61, 80].

For corrosion protection, it is relevant to evaluate the evolution of the WCA after a prolonged exposition to the aqueous solution. As shown in Figure 12a, DSS2205 bare surfaces are hydrophobic, with WCA values lower than those of LST surfaces, as expected given its smoother surface. Moreover, WCA is almost indifferent with respect to samples analyzed before (105.3±1.5°) and after (97.3±8.2°) potentiodynamic tests (noncyclic). In the case of LST surfaces, the superhydrophobicity (WCA 153.9±2.3°) decreases

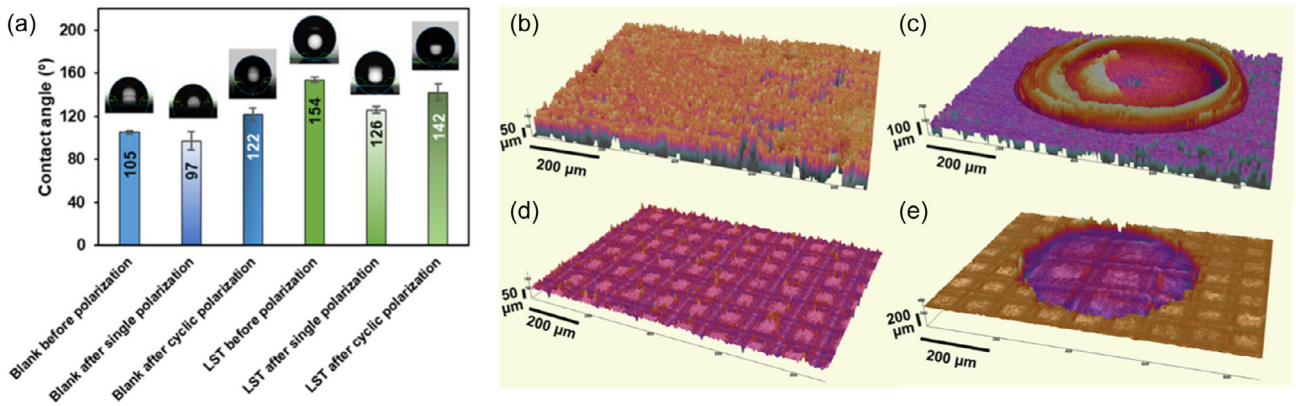


FIGURE 12 | (a) WCA degrees for DSS2205 surfaces, corresponding to control (Blank) and LST samples, before and after 7 days of potentiodynamic polarization tests (noncyclic) and cyclic potentiodynamic polarization. (b,c) 3D surface topography of Blank top side, before and after the water drop deposition, respectively. (d,e) 3D surface topography of LST surface, before and after the water drop deposition, respectively. Both samples were subjected to the cyclic potentiodynamic polarization assays during 1 week.

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by 18% after 1 week of polarization analysis (125.9±3.3°); however, the interface metal–water is still highly hydrophobic thanks to its micropatterned structure. After cyclic polarization experiments, both the laser-treated and the bare samples, showed increased hydrophobicity (142.3±8.0° and 121.9±5.9°, respectively), apparently due to the presence of particles detached from the base metal and the micropitting observed by SEM mapping analysis. On the contrary, the LST samples are still much more hydrophobic than nontreated samples, which justifies the fast pitting corrosion observed in Blank substrates in NaCl solution and with cyclic potentiodynamic experiments.

In order to evaluate the influence of grooves and pillar in the droplet deposition, 3D surface topography images were taken before and after the contact with the water and compared with Blank surfaces. As can be seen in Figure 12b,c, the drop of water intrinsically spreads across the entire smooth surface of DSS2205 steel, supposedly due to the presence of metal oxides. Figure 12d shows the 3D topography of laser textured surface, where the pattern composed by microchannels and pillars can be distinguished. Further, Figure 12e clearly evidences the water drop spread either in pillars or in grooves regions. Even having more roughness than Blank, the wettability was decreased in such surface. Some authors have attributed this hydrophobic effect to a stable state created after a period of time from laser treatment, thanks to the stable adsorption of oxygen and carbon compounds from air [61]. However, there are also other theories which were not fully demonstrated until now.

4 | Conclusions

The stability of DSSl 2205 modified by LST in brine solutions, after cyclic electrochemical tests, has been satisfactorily evaluated. Although flat and nontextured surfaces of DSS2205 are considered one of the highest resistant steels in marine environments, ferrite seems to be dissolved in prolonged electrochemical cyclic experiments, causing the appearance of micropits, as proved in the present study. By the contrary, the LST alloy maintains its patterning of grooves and pillars designed by the nanosecond-laser treatment, even after the severe damage caused by 1 week of brine solution immersion and cyclic potentiodynamic testing, preserving its high hydrophobicity properties and avoiding crevice corrosion. Therefore, the present study demonstrated that laser-texturing treatments, with controlled laser fluence and scanning speed, generate a stable oxide passive film on the DSS2205 top face, which enhances the stability of stainless steel matrix in brine solution (3.5 wt.%) and under harsh conditions.

Evaluation of the surface topography and corrosion products by FTIR, Raman, SEM–EDX and XRD analyses corroborated the abovementioned results. The samples without laser texturing treatment had evident signs of crevice localized corrosion, whereas LST samples maintained the patterned grooves intact. This protective effect is attributed to the obtaining of $FeCr_2O_4$ oxides in the passive film localized at the grooves, which was induced by the LST process. Therefore, a continuous overlap of the laser tracks in LST treatment is necessary to achieve such good performance.

One of the most important conclusions is that flat and SLT DDS2205 stainless steel cannot be recommended in systems which will involve immersion of the pieces in NaCl solutions with cyclic electrochemical procedures, such as in electrolysis pumps for generating chlorine gas in swimming pools, in osmosis desalination plants, among other sectors. However, SLT DDS2205 can be employed in non-submerged zones with chlorine atmosphere (ports, ships, shipyards, coastal areas) under continuous monitoring (maintenance) because the passive oxide layers are more stable than untextured ones.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Supplementary information and research data sets can be provided upon authors request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. Additional FTIR spectrum, SEM-EDX spectra with mapping analysis, optical micrographs, potentiodynamic polarization curves and tables have been included in the supporting file (PDF). **Supporting Fig. S1**: FTIR spectra of DSS2205 surfaces with the main functional groups for observed inLST samples. Blank sample did not accuse any absorption band in transmittance mode. **Supporting Fig. S2**: EDX spectra corresponding to Blank sample: a) before and b) after single polarization test in NaCl 3.5 wt.%. **Supporting Fig. S3**: EDX spectra corresponding to LST sample: a) before and b) after single polarization test in NaCl 3.5 wt.%. **Supporting Fig. S4**: Cyclic potentiodynamic polarization curves of: a) Blank sample and b) SLT sample, with increasing immersion time in brine solution (NaCl 3.5wt. %). **Supporting Fig. S5**: Optical micrographs of samples surfaces after the cyclic tests: (a, b) Blank (5× and 20× magnifications, respectively); and (c-d) SLT (5× and 20× magnifications, respectively). **Supporting Fig. S6**: (a) SEM high magnification image of pillar zone highlighting the presence of pits. (b) EDX mapping images of main 2205 DSS elements and alloying atoms (Fe, O, Mn, Ni, Cr, C, Mo). (c) EDX spectrum of SEM image showed in a), where NaCl atoms are also present together with the alloy composition. **Supporting Table S1**: EDX atomic weight percentage (%) of alloying elements, and Cr/Fe and Cr/Ni ratios detected at the Blank and LST surfaces, before and after 7 days of potentiodynamic anodic polarizations. **Supporting Table S2**: Characteristic parameters acquired from the cyclic potentiodynamic polarization curves with increasing immersion time in NaCl solution (3.5wt. %). **Supporting Table S3**: EDX atomic weight percentage (%) of alloying elements and Cr/Fe, Cr/Ni ratios detected at the Blank and SLT surfaces, before and after 7 days of cyclic potentiodynamic polarizations in NaCl 3.5 wt.%.

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